Chapter 1 · Energy Causes Change

GUIDED NOTES

Name: _____ Homeroom: ○ 4A ○ 4B ○ 4C ○ 4D

BQ Where can we observe evidence of energy causing change?

My first thinking

A pencil gets shorter and sharper.

V1 VOCABULARY — ENERGY & CHANGE

Reader p. 1

energy — the ability to cause **change** .

change — to become **different** .

N1 ENERGY CAUSES CHANGE

Reader pp. 1–2

Energy has the ability to cause change.

When you see a change, you are seeing **evidence** of energy.

ENERGY	→	CHANGE	→	EVIDENCE

N2 ENERGY OCCURS IN DIFFERENT FORMS

Reader p. 3

Write one example that shows a change for each form.

FORM	ONE EXAMPLE THAT SHOWS A CHANGE
motion	a ball rolls across the floor
sound	a drum makes a noise
light	a flashlight lights up the room
heat	a warm mug warms my hands
electrical energy	a fan spins when plugged in

Sample answer — student wording may vary.

Chapter 1 · Energy Causes Change

V2 VOCABULARY — STORED ENERGY

Reader p. 4

stored energy — energy that has the ability to cause change at a **later** time.

N3 OBJECTS CAN STORE ENERGY

Reader pp. 4–5

STORED ENERGY	→	RELEASE	→	CHANGE
battery (in a flashlight)		the flashlight lights up		
stretched rubber band		it snaps and makes a sound		
raised object (apple)		it falls down		

Objects also have stored energy because of their **position**.

N4 RECOGNIZING EVIDENCE

Reader pp. 1–5

Draw a change you observed. Show what it looked like BEFORE and AFTER.

BEFORE	AFTER
Sample: a dull pencil, long, rounded tip	Sample: shorter pencil, sharp tip, shavings

What changed?

The pencil got shorter and sharper.

CHECK YOUR THINKING

Reader pp. 1–5

How do we know energy is involved if we cannot see energy itself?

Start with: “We know because we see …”

We know because we see a change, like a ball starting to move

or a light turning on.

BQ · NOW I THINK Where can we observe evidence of energy causing change?

We see evidence of energy when things move, make sound, light up,

heat up, or change shape.

Sample answer — student wording may vary.